

Rethinking Literature Search Evaluation: Deep Research Helps, and Human Citation Lists Are Not a Ground Truth

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Abstract

We study large-scale literature search from two complementary angles: improving the retrieval pipeline, and stress-testing the human reference list as an evaluation target. First, we implement a *Deep Research* pipeline that processes the full query paper and expands the retrieved results breadth-first along their bibliographies, and show that it substantially outperforms vanilla API-only search, raising recall on ROLLINGEVAL-JUN25 (a 250-paper literature-search benchmark) from below 20% to above 80%. Second, we use a neutral LLM-as-a-judge to determine if human references are sound ground truth for the task. We find significant limitations: only 51% of human citations are judged moderately relevant or higher, against 86–88% for the strongest AI-based re-rankers. We study this gap on the OpenAlex co-authorship graph, finding that humans are $2.5\times$ more likely than the best AI re-rankers to cite a direct collaborator. Together, our results argue against single-axis literature-search evaluation: recall, topical-relevance scoring, ranked-list diversity, and a co-authorship-distance diagnostic each measure complementary properties of citation quality and should be reported jointly.

1 Introduction

Literature search remains a foundational step of scientific work, yet it is hard to evaluate well. The community typically reports precision and recall against the papers that the query paper itself cites (Ajith et al., 2024; Hang et al., 2025), treating that reference list as ground truth. This convention has two problems. First, vanilla API search based on keywords misses most of the relevant literature: single and multiple keyword queries to a scholarly index are brittle, and the long-tail structure of citation graphs hides relevant work behind chains of references (Kinney et al., 2023; Priem et al., 2022). Second, the reference list itself is incomplete and idiosyncratic (Agarwal et al., 2025), including foun-

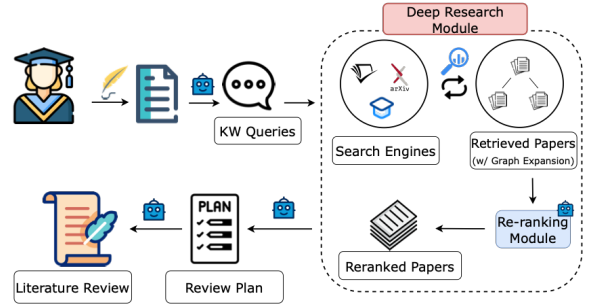


Figure 1: Deep Research pipeline. The system ingests the full query paper, generates LLM-constructed keyword queries, retrieves seed papers from scholarly APIs, expands the candidate set along citation links, and re-ranks the pooled candidates. Full algorithm and prompts in Appendix B.

dational paper citations and software attribution, in addition to topically relevant papers.

We address both problems and report two findings. (1) A *Deep Research* pipeline that ingests the full query paper and recursively expands its bibliography increases recall by an order of magnitude over vanilla API search. (2) Treating human reference list as a clean ground truth is problematic: a neutral LLM judge (Zheng et al., 2023) rates only 51% of human citations as moderately relevant or higher, compared with 86–88% for the strongest automated re-rankers. We probe this gap through the OpenAlex co-authorship graph (Priem et al., 2022) and find that humans cite direct collaborators $2.5\times$ more often than the strongest re-rankers.

Contributions. (i) A modular Deep Research retrieval pipeline (Figure 1) for full-text literature search, with order-of-magnitude recall gains over traditional literature search baselines, along with ROLLINGEVAL-JUN25, a 250-paper benchmark that mitigates data contamination. (ii) An analysis of human reference lists that documents systematic non-topical citing and provides a framework for re-thinking literature-search ground truth.

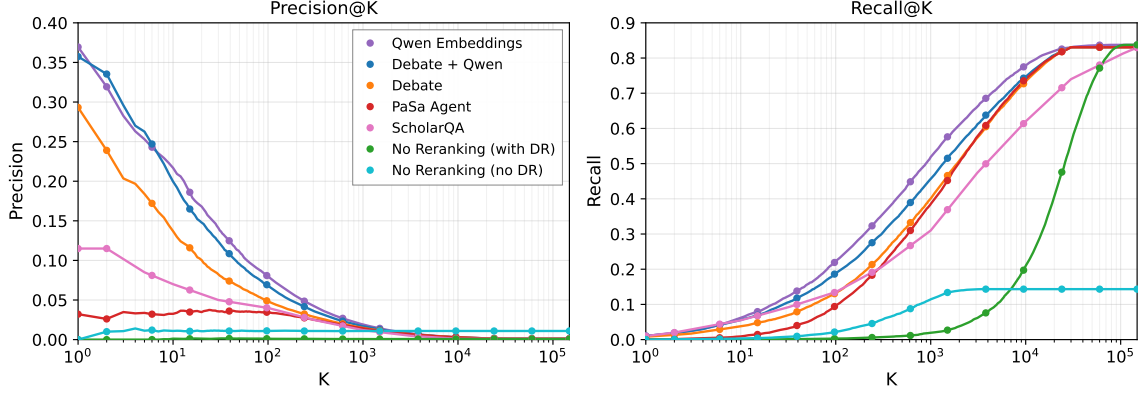


Figure 2: Precision@ K (left) and Recall@ K (right), K on log scale. Deep Research raises recall by an order of magnitude over normal search; QWEN3 embeddings give the strongest top- K precision.

2 Deep Research Pipeline

The system has two phases: a high-recall retrieval phase and a re-ranking phase. **Phase 1** prompts an LLM to draft diverse keyword queries for the query document, translates them into provider-specific syntax for arXiv, OpenAlex, and Semantic Scholar (Kinney et al., 2023; Priem et al., 2022), and collects candidates under rate-limit-aware concurrency. **Phase 2** (Deep Research) expands the candidate set breadth-first along bibliographic references up to a configurable depth and paper budget. One of three interchangeable modules then re-ranks the expanded pool: QWEN3 embeddings (Zhang et al., 2025a); an LLM-based debate ranking mechanism that produces arguments for and against inclusion before assigning a 0–100 relevance score; or a Debate+QWEN3 ensemble that combines the two. Algorithm pseudo-code, prompts, and full caching/concurrency details are in Appendix B.

3 Setup

Benchmark. We construct ROLLINGEVAL-JUN25, a June 2025 snapshot of the ROLLINGEVAL series proposed originally by (Agarwal et al., 2025) to avoid data contamination issues when benchmarking LLMs for literature search. The benchmark comprises 250 computer science arXiv submissions with cleaned full-text and bibliographies. We choose June 2025 as it post-dates the training-data cutoff of every LLM evaluated in this work, mitigating contamination, and the cleaned full-text and bibliography pipeline supports the matched-input head-to-head re-ranker comparison in Section 4. We fetch references through the OpenAlex and SemanticScholar APIs, and fall back on an LLM-based parser that extracts

bibliographies directly from PDFs.

Metrics. We report: (a) Precision and Recall against the human reference list, (b) Semantic Relevance (SR), a 0–100 score from an LLM judge (Zheng et al., 2023), and (c) α -nDCG (Clarke et al., 2008), the novelty-aware variant of nDCG (Järvelin and Kekäläinen, 2002) that measures how well a ranked list covers diverse topics without redundancy. Detailed metric definitions are in Appendix D.

The SR judge is GPT-OSS-120B served via vLLM (Kwon et al., 2023); for each pair it conditions on the query paper’s full text (title, abstract, body) and the candidate’s title and abstract, and returns a rubric grade in $\{0, \dots, 5\}$ (see Prompt 8), which we then multiply by 20. The re-rankers also score candidates by comparing the query paper’s full text with the candidate’s title and abstract.

4 Retrieval Results

Deep Research yields an order-of-magnitude recall gain. Figure 2 plots precision and recall against the human reference list. Vanilla API search (*No Rerank, no DR*) tops out near 15% recall only at $K \approx 10^5$ and never exceeds it, while every Deep Research variant exceeds 80% recall by $K=10^4$. QWEN3 embeddings achieve the strongest top- K precision (0.37 at $K=1$), closely followed by the Debate+Qwen ensemble. We also include PASA (He et al., 2025) and SCHOLARQA (Singh et al., 2025) as deep-research baselines. We run both on ROLLINGEVAL-JUN25 with the query paper’s abstract as input; each system then produces its ranking through its native retrieval and scoring pipeline. Table 1 summarizes precision and recall at four representative cutoffs. The best Deep

Method	Precision			Recall		
	@20	@100	@1k	@20	@100	@1k
No Rerank (no DR)	0.1	0.1	0.0	0.0	0.3	1.9
PASA	5.3	4.5	1.5	1.3	7.4	27.8
SCHOLARQA	5.7	4.0	1.2	7.7	13.5	31.0
Debate	9.6	4.5	1.5	5.1	12.2	39.9
Debate+Qwen3	14.7	6.6	1.7	8.1	18.1	46.2
QWEN3 emb.	16.8	8.0	1.9	9.6	22.2	51.9

Table 1: Precision and recall at $K \in \{20, 100, 1,000\}$. Best column in bold.

Method	α -nDCG			Sem. Rel.		
	@10	@100	@1k	@10	@100	@1k
No Rerank (no DR)	15.3	41.2	62.1	9.2	17.3	17.4
Debate+QWEN3	53.2	76.2	87.9	67.9	53.8	33.2
Debate	54.1	79.7	89.1	55.4	39.8	27.1
QWEN3 emb.	47.6	65.3	77.8	67.1	57.6	47.9
Human References	—	—	—	47.3	50.7	50.8

Table 2: α -nDCG and Semantic Relevance at $K \in \{10, 100, 1000\}$. Higher is better.

Research variant nearly triples recall at $K=100$ over an arXiv-API baseline and reaches 51.9% recall at $K=1000$.

Diversity and ranking quality. Beyond precision and recall, we ask whether highly relevant items are concentrated near the top of each ranked list. Table 2 reports α -nDCG (Clarke et al., 2008) and the LLM-judged Semantic Relevance score at $K \in \{10, 100, 1000\}$. Debate (without Qwen) achieves the strongest α -nDCG at $K=100$ and $K=1000$, suggesting that prompt-based scoring leads to higher diversity and relevant items more evenly through the ranked list than embedding similarity. The Debate+QWEN3 ensemble has the highest top-10 Semantic Relevance (67.9), validating it as the strongest top-of-list retriever, while QWEN3 alone maintains the highest Semantic Relevance at $K=100$ and $K=1000$, indicating better long-tail behavior. The ground-truth row anchors the comparison: the human reference list itself sits at 47–51 on Semantic Relevance across K , well below the AI re-rankers in the top-10/100 buckets, a discrepancy we examine next.

5 Human References = Ground Truth?

Recall against the human reference list is a coherent metric only if that list is itself reliable. We test this by asking a neutral LLM judge to rate the topical relevance of every cited paper, then comparing the resulting distribution against the same

Source	Mean	≤ 40	≥ 60	n
Human	51.2	48.6%	51.4%	9,204
QWEN3 emb.	62.4	14.0%	86.0%	9,173
Debate+QWEN3	62.4	12.2%	87.8%	9,173
Debate	45.6	37.9%	62.1%	9,173
No Rerank (no DR)	17.3	97.7%	2.3%	9,173

Table 3: Semantic relevance distribution at matched top- N_P . n is the total number of (query, reference) pairs.

judge’s ratings of AI re-ranker outputs at matched list length.

Setup. For each of the 250 query papers P with N_P human references, we evaluate all human citations and the top- N_P predictions from each re-ranker. This list-length parity is essential: a longer AI list would trivially include more low-scoring tail items. We aggregate 9,204 human (query, reference) pairs and 36,692 AI pairs.

Table 3 shows the matched- K distribution and Figure 3 the full curve. Restricted to the same list length, human references score ≥ 60 in 51.4% of cases, while QWEN3 and Debate+Qwen reach 86.0% and 87.8%. The disparity concentrates at score = 40, which captures 38.2% of human references but only 10.2–13.7% of what the top re-rankers retrieve. On the LLM judge’s topical-relevance axis, the strongest re-rankers’ top- N_P lists score higher than the human-curated list at matched length; this speaks to one axis of citation quality, not an overall verdict on the human list. Figure 3 makes this concrete across K : the AI re-rankers start well above the human saturation level and remain above it throughout the $K \leq 1000$ scoring window, with their cumulative scores gradually diluting toward the matched- K means in Table 3 as the top-of-list signal blends with lower-relevance tail items.

Is the gap a citation-collusion artifact? A natural hypothesis is that humans cite collaborators. We test this on the OpenAlex co-authorship graph (Priem et al., 2022), classifying each (query, candidate) pair by the minimum hop distance between author sets, $d \in \{0, 1, 2, 3, \geq 4\}$ (formal definition and BFS data requirements in Appendix E). Table 4 reports the distribution.

The bias concentrates at short hops and decays with radius. At $d=0$ (direct co-authorship) humans are $2.5\times$ more likely than the strongest re-rankers to cite themselves or a co-author (5.13% vs. 1.9–2.1%). The cumulative within-one-hop rate is 12.2% for humans vs. 6.9–7.4% for the best AI

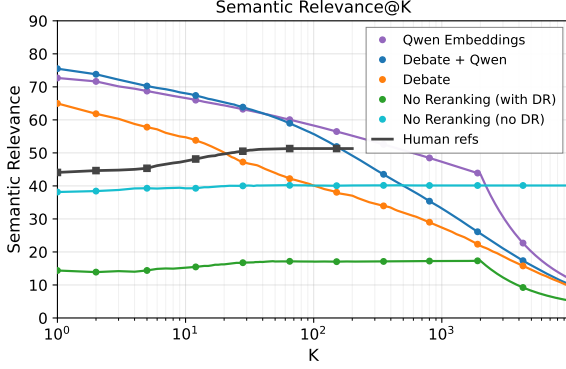


Figure 3: Cumulative semantic relevance over the top- K retrieved candidates, with the SR judge applied to at most $K_{\max}=1000$ items per query. Human references saturate near 51 by $K=50$; AI re-rankers start substantially higher (73–75 at $K=1$) and stay above the human curve across the full $K \leq 1000$ scoring window.

Source	$d=0$	$d=1$	$d=2$	$d=3$	$d \geq 4$
Human	5.13	7.02	41.43	29.13	17.28
QWEN3	2.05	5.36	41.25	34.14	17.19
Debate+QWEN3	1.92	4.98	39.69	36.06	17.35
Debate	1.52	4.32	38.18	36.72	19.27
No Rerank (no DR)	0.09	2.65	36.87	40.17	20.22

Table 4: Co-authorship-graph distance at matched top- N_P , OpenAlex graph. $d=0$: direct co-authorship; higher d : shortest co-author path.

methods, a $1.6\times$ gap. By $d \leq 2$ the gap narrows to $1.1\times$ (53.6% vs. 46.6–48.7%), and by $d \leq 3$ all sources converge near 80–83%, consistent with the small-world structure of the academic graph. Conditional mean scores are essentially flat across distance classes for humans (56.1, 52.6, 51.2, 51.2, 50.9 for $d=0$ through $d \geq 4$): the network proximity of a citation does not predict its topical relevance.

Interpretation. Both effects are individually defensible but together explain the gap. The network bias is concentrated at direct and one-hop collaborators, where humans diverge sharply from the re-rankers (5.13% vs. 1.9–2.1% at $d=0$; 12.2% vs. 6.9–7.4% within one hop); from $d \geq 2$ onward the distributions converge, so network proximity drives a clean but bounded share of the disparity. Citing close collaborators is appropriate where the work is genuinely relevant, but at this magnitude it suggests authors should put proportionally more effort into capturing the broader landscape beyond their immediate clique. On the topical axis, foundational background (Transformer, ResNet, BERT (Vaswani et al., 2017; He et al., 2016; Devlin et al.,

2019)), classical algorithms (Adam (Kingma and Ba, 2015)), and software or library attribution are all defensible reasons to cite a paper whose abstract carries no topical signal; what is harder to defend is that citations of this kind make up nearly half (48.6%) of human reference lists at matched length, against 12–14% for the strongest re-rankers. Eighteen representative examples are in Appendix F. The practical implication is two-fold: literature-search benchmarks should not treat human reference lists as a clean ground truth, and authors should treat the topical-relevance share of their bibliography as a property worth checking. Future work should also pair the topical rubric with citation-context spans from the query paper, or otherwise model the reasons humans cite.

6 Discussion and Conclusion

Our results argue for separating two questions the community usually conflates. *Did the system retrieve the papers the authors cited?* is a coverage question against a known-imperfect target. *Did the system retrieve papers that a neutral reader would judge relevant?* is a different, complementary question. The two diverge sharply: Deep Research dominates on the first, while the human-curated list scores lower than the top re-rankers on the second. These two findings are intertwined rather than competing: the order-of-magnitude recall gain matters precisely because it expands the candidate pool from which any retrieval system, including SR-aligned ones, must draw, so a higher-recall pipeline raises the achievable ceiling on whichever metric the community ultimately adopts as ground truth. Neither metric is strictly better: recall against human citations captures dimensions an LLM-judge cannot justify (foundational and tool/library attribution, methodological lineage), while the topical rubric captures dimensions that human reference lists undersample (long-tail topically-related work, non-collaborator authors). We argue for reporting both, alongside the OpenAlex-graph diagnostic, rather than treating either in isolation as ground truth. We will release ROLLINGEVAL-JUN25, the Deep Research pipeline, the LLM-judge prompts, and the OpenAlex distance analysis as a reproducible scaffold for re-examining literature-search ground truth.

Limitations

The LLM judge is a single neutral model; rubric-conditioned scoring may drift across model and prompt versions, and we do not currently model citation context (in-text spans surrounding the citation). Validating against human relevance ratings on a labelled subsample remains a natural next step. The co-authorship graph fetch caps each author at 1,000 most recent works, so we under-count distances through hyper-prolific intermediaries, which biases the reported gap in a conservative direction. Bibliography extraction relies on LLM agents and can carry residual noise. The ROLLINGEVAL benchmark consists only of computer science papers; extending it to biomedical, social-science, and humanities papers would test cross-domain generality. Finally, the analysis covers only arXiv-indexed references; we exclude non-arXiv citations.

Potential Risks

The Semantic Relevance judge and Deep Research pipeline we release target legitimate literature discovery, but the same topical-relevance rubric could be misapplied to gate citations during peer review, to filter submissions, or to automate rejection, uses for which a single LLM judge is not a defensible standard. Our results show the rubric and the human reference list disagree systematically along measurable axes; both should serve as complementary diagnostics, not as evaluation thresholds. The OpenAlex co-authorship analysis surfaces network proximity between authors that is computable from public data but that researchers may reasonably consider private; we report only aggregate distributions and avoid disclosing individual-paper proximities. Finally, ROLLINGEVAL-JUN25 comprises arXiv preprints chosen to postdate the training cut-off of every LLM we evaluate; its contamination-resistance guarantee weakens as models update, so the snapshot will need periodic refreshes. Additionally, extending it to include articles from other domains medical, biology, or physics might help study its generalizability.

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A Background and Related Work

Scholarly APIs and index-based retrieval.

Scholarly APIs such as OpenAlex (Priem et al., 2022), Semantic Scholar (Kinney et al., 2023), and arXiv remain the backbone of scalable literature discovery, but require careful query design to maximize recall under strict rate limits, and a single API call against them typically saturates well below the topical neighborhood of the query paper. We therefore use these APIs only as the entry-point fetch layer, drafting an LLM-constructed keyword set against them in Phase 1 and then expanding breadth-first along bibliographic references in Phase 2 (Section 2).

Dense retrievers and re-rankers. Dense retrievers such as DPR (Karpukhin et al., 2020) and BM25 (Robertson and Zaragoza, 2009), together with citation-aware embeddings like SPECTER (Cohan et al., 2020), provide stronger semantic matches than keyword search but trade off efficiency at scale; more recent LLM-guided schemes such as SemRank (Zhang et al., 2025b) graft concept-level matching on top of a dense retriever. We treat the re-ranker as an interchangeable module: QWEN3-EMBEDDING-8B (Zhang et al., 2025a), an LLM-based Debate ranker, and their ensemble plug into the same final stage of the pipeline and are compared head-to-head under matched inputs and matched candidate pools.

LLM-based retrieval agents. LLM-based retrieval agents such as PaSa (He et al., 2025), Ai2 ScholarQA (Singh et al., 2025), PaperQA (Lála et al., 2023), RhinoInsight (Lei et al., 2025), and earlier LitLLM (Agarwal et al., 2025) combine API search with LLM scoring, control mechanisms, or bibliography expansion, and Zhang et al. (2026) extend the family to a multi-turn, tool-using policy trained with reinforcement learning. We hold the orchestration policy constant (keyword expansion followed by bibliography-graph expansion) and instead vary (i) full-text input vs. abstract-only input and (ii) the choice of re-ranker, which isolates the contribution of full-paper context and citation-graph reach from the choice of agent policy.

Evaluating LLM literature search. On the evaluation side, the LITSEARCH benchmark (Ajith

et al., 2024) and the three-metric framework of Hang et al. (2025), which scores GPT-4, Gemini 2.5, and DeepSeek-V3 in vanilla and “deep research” configurations, both anchor quality on recall against the human reference list. Our analysis treats that recall as one diagnostic axis, pairs it with a semantic-relevance rubric scored by a neutral LLM judge, and adds an OpenAlex co-authorship-graph distance metric (Section 5); the first two metrics disagree systematically on real data, and the co-authorship analysis attributes a measurable portion of the gap to network bias in human citing rather than to retrieval quality alone.

B System Details

B.1 Phase 1: Retrieval with LLM-Constructed Queries

Given a query document, we prompt an LLM to draft a diverse keyword query set, then normalize each query into the provider-specific syntax for arXiv, OpenAlex, and Semantic Scholar (Kinney et al., 2023; Priem et al., 2022). We pre-process inputs to remove inline citations and bibliography content before prompting, and cache all generated queries by input- and model-specific hash.

B.2 Phase 2: Citation Graph Expansion

We expand the seed set breadth-first along bibliographic references up to a configurable depth D_{max} ($=3$ in our experiments) and paper budget N_{max} . When references from APIs like Semantic Scholar and OpenAlex are incomplete, we fall back on LLM-based PDF bibliography parsing, which first extracts the bibliography section from the paper and resolves each entry through its title and authors using Semantic Scholar and OpenAlex. Algorithm 1 states the procedure.

B.3 Re-ranking Modules

- QWEN3 embeddings (Zhang et al., 2025a) served via vLLM (Kwon et al., 2023), scoring each candidate by cosine similarity between the instruction-conditioned query embedding and the bare candidate embedding.
- LLM debate ranking: an LLM agent scores each candidate (0–100) with arguments-for/against conditioned on the query paper’s full text. Batches run asynchronously under a concurrency semaphore.

Algorithm 1 Deep Research: Citation Graph Expansion

Require: Initial Seed Papers P_{seed} , Max Depth

D_{max} , Max Papers N_{max}

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1:  $Q \leftarrow \text{new Queue}()$ 
2:  $P_{final} \leftarrow P_{seed}$ 
3:  $V_{ids} \leftarrow \{\text{id}(p) \text{ for } p \in P_{seed}\}$ 
4: for each  $p \in P_{seed}$  do
5:    $Q.\text{enqueue}((p, 0))$ 
6: end for
7: while  $Q$  is not empty and  $|P_{final}| < N_{max}$  do
8:    $(p_{curr}, d_{curr}) \leftarrow Q.\text{dequeue}()$ 
9:   if  $d_{curr} \geq D_{max}$  then
10:    continue
11:   end if
12:    $P_{cand} \leftarrow \text{GetReferences}(p_{curr})$ 
13:   for each  $p_{new} \in P_{cand}$  do
14:     if  $\text{id}(p_{new}) \notin V_{ids}$  then
15:        $V_{ids}.\text{add}(\text{id}(p_{new}))$ 
16:        $P_{final}.\text{add}(p_{new})$ 
17:        $Q.\text{enqueue}((p_{new}, d_{curr} + 1))$ 
18:       if  $|P_{final}| \geq N_{max}$  then
19:         break
20:       end if
21:     end if
22:   end for
23: end while
24: return  $P_{final}$ 

```

- Debate+QWEN3 ensemble: averages the two normalized scores into a single per-candidate ranking.

B.4 Engineering Notes

All experiments use asynchronous execution with bounded parallelism, query/result caching keyed by input hash, and YAML-driven configuration. API semaphores respect provider limits while maintaining throughput.

C Prompt Templates

This appendix lists the four prompts the Deep Research pipeline and LLM-as-a-judge evaluation use: (i) keyword generation for Phase 1 retrieval (Prompt 5), (ii) LLM debate ranking (Prompt 6), (iii) the instruction prefix for QWEN3 embedding queries (Prompt 7), and (iv) the semantic-relevance judge (Prompt 8) that produces all scores in Sections 4–5.

C.1 Keyword Generation

Phase 1 prompts an LLM to draft a diverse set of search queries given the query paper’s abstract.

C.2 Debate Ranking

The debate re-ranker (Table 6) scores each candidate on a 0–100 scale by first writing arguments for and against citing it. We batch candidates per LLM call and share the system prompt with the keyword-generation step.

C.3 QWEN3 Embedding Instruction

Following the recommended usage of QWEN3-EMBEDDING-8B (Zhang et al., 2025a), we prefix each query paper with a one-sentence task instruction before embedding, and embed candidate abstracts without one. Cosine similarity between instruction-conditioned query embeddings and bare candidate embeddings produces the final ranking.

C.4 Semantic Relevance Judge

Every Semantic Relevance score we report (the matched- K distribution in Table 3, the cumulative curves in Figure 3, and the per-method scores in Table 2) comes from the prompt in Table 8. We show the judge the query paper (title, abstract, optionally full text) and a single candidate (title, abstract); it returns an integer in $\{0, 1, 2, 3, 4, 5\}$, which we multiply by 20 to obtain the 0–100 scale used in the main text.

D Metrics: Definitions

Precision and Recall. For retrieved set $R@K$ and ground-truth set G ,

$$\text{Precision@K} = \frac{|R@K \cap G|}{|R@K|}$$

$$\text{Recall@K} = \frac{|R@K \cap G|}{|G|}$$

We report curves for K up to 1.5×10^5 to assess long-tail retrieval.

LLM-as-a-judge relevance. For each query–candidate pair, an LLM (Zheng et al., 2023) assigns a graded relevance score on a 5-point rubric, which we multiply by 20 to obtain a 0–100 scale. The judge conditions on the query paper and the candidate’s title and abstract.

α -nDCG. We extend nDCG (Järvelin and Kekäläinen, 2002) with the novelty-aware variant of Clarke et al. (2008). For each retrieved document d that belongs to a ground-truth cluster c , the contribution at rank r is

$$\text{gain}(d, r) = \frac{1}{\log_2(r + 1)} \cdot (1 - \alpha) \alpha^{n_c},$$

where n_c is the number of earlier documents from cluster c . We normalize by the ideal gain to obtain α -nDCG@K. Clusters come from HDBSCAN (McInnes et al., 2017) over Qwen3 embeddings (Zhang et al., 2025a) of candidate-paper abstracts, with the default `min_cluster_size`, so cluster identity reflects topical neighborhoods of the candidate pool.

E Co-Authorship Graph Distance: Definition and Computation

This appendix formalizes the co-authorship-graph distance metric from Section 5 and documents the data we fetch to compute it.

Graph and distance. Let $G = (V, E)$ be the undirected co-authorship graph in which V is the set of academic authors and an edge $(a, b) \in E$ exists whenever a and b have co-authored at least one publication. For a query paper Q with author set A_Q and a candidate paper C with author set A_C , the pair distance is

$$d(Q, C) = \min_{a \in A_Q, b \in A_C} d_G(a, b), \quad (1)$$

where $d_G(\cdot, \cdot)$ is the standard shortest-path distance on G , with $d_G(a, a) = 0$ and $d_G(a, b) = \infty$ for disconnected pairs.

Role	Content
System	You are a helpful research assistant who is helping with literature review of a research idea.
User	You are a helpful research assistant assisting with a literature review for a research idea. You will be given the abstract of a scientific paper. Your goal is to generate a diverse set of mutually exclusive search queries to help find relevant and citable academic papers using scholarly search engines. Here is the abstract: {paper_text} Instructions. • Generate 10 search queries written in the natural, concise style typically used by researchers using academic search engines (e.g., OpenAlex, Semantic Scholar, or Google Scholar). • Each query should reflect a different angle of the abstract (e.g., method, task, dataset, domain, application, or novelty). • Use a variety of keywords and phrasings to maximize diversity and avoid redundancy. • Focus on maximizing recall (retrieving a broad but relevant set of papers), not just precision. • Avoid stopwords or overly verbose phrasing; use terms that researchers would actually search for. • Return a JSON object with the following structure: <pre>{ "queries": ["first query here", "second query here"] }</pre>

Table 5: Keyword generation prompt (Phase 1 of Deep Research).

Hop neighborhoods. For a set $S \subseteq V$,

$$L_0(S) = S,$$

$$L_{k+1}(S) = \bigcup_{u \in L_k(S)} N(u),$$

where $N(u) = \{v : (u, v) \in E\}$ is the set of co-authors of u .

Distance test. We determine the pair distance by checking each level in increasing order:

$$d=0 : A_Q \cap A_C \neq \emptyset,$$

$$d=1 : L_1(A_Q) \cap A_C \neq \emptyset,$$

$$d=2 : (L_1(A_Q) \cap L_1(A_C) \neq \emptyset) \vee (L_2(A_Q) \cap A_C \neq \emptyset),$$

$$d=3 : L_2(A_Q) \cap L_1(A_C) \neq \emptyset.$$

We record pairs failing all four tests as $d \geq 4$. The $d=2$ disjunction is necessary because a length-2 path takes either form: a shared collaborator, or a second-hop reach.

Data requirements. We materialize three subsets:

- $L_1(A_Q)$ requires the co-author list of each of the 1,113 unique query authors.
- $L_1(A_C)$ requires the co-author list of each candidate author that is not already in

$L_1(A_Q) \cup A_Q$ (52,868 authors after exclusion).

- $L_2(A_Q)$ requires the co-author list of each author in $L_1(A_Q)$ (114,669 authors), which is the dominant cost.

$d=3$ detection requires *both* the candidate-side fetch and the second-hop expansion: the $d=3$ intersection $L_2(A_Q) \cap L_1(A_C)$ draws one factor from each side.

Bounded-degree approximation. We construct each author’s co-author list by aggregating across the OpenAlex works in 5 pages of 200 results. Under-counting can only inflate the estimated distance, so the reported gap is a conservative lower bound.

Identifier reconciliation. We resolve author identities through OpenAlex identifiers (Priem et al., 2022), and map arXiv papers to OpenAlex Works via the batched DOI filter using the prefix 10.48550/arxiv.. Of 22,935 unique arXiv identifiers, 19,990 (87.2%) resolve; we exclude pairs in which either side fails.

Cost. OpenAlex’s batched author-id filter accepts up to 50 IDs per request at \$0.0001 per request under the prepaid plan. The complete pipeline costs approximately \$1.33 in API spend and 4–5 hours of wall-clock time with parallel fetches.

Role	Content
System	You are a helpful research assistant who is helping with literature review of a research idea.
User	You are a helpful research assistant. Your task is to rank some papers based on their relevance to a query paper. Given the query paper: <pre><query_paper> {query_full} </query_paper></pre> And the following candidate reference paper abstracts: <pre><candidate_paper_abstracts> {reference_papers} </candidate_paper_abstracts></pre> Instructions. • For EVERY candidate paper, provide a relevance score between 0 and 100 representing the probability that the query paper would cite it. • The relevance score MUST be an integer between 0 and 100 (inclusive). • The score MUST be written using digits only (e.g., 0, 17, 42, 50, 100). • Do NOT write numbers in words (e.g., “thirty-seven” or “fifty”). • If a candidate paper happens to be a duplicate of the query paper, it should receive a score of 0. • Provide arguments for and against citing the candidate paper, extracting supporting sentences from the candidate’s abstract. • Format your response for EACH paper using the specified tags below. Response format for each paper: <pre><arguments_for> [paper’s id]: [Arguments for including the paper] Extracted Sentences: "Sentence 1", "Sentence 2", . . . </arguments_for> <arguments_against> [paper’s id]: [Arguments for not including the paper] Extracted Sentences: "Sentence 1", "Sentence 2", . . . </arguments_against> <probability> paper_id: [paper’s id] score: [Final Probability Score Based on the Arguments]/100 </probability></pre> Note: your response MUST contain arguments and probabilities for ALL the candidate paper abstracts.

Table 6: Debate ranking prompt. Arguments-for/against and a 0–100 score are produced per candidate.

Side	Content
Query pre-fix	Instruct: You are a helpful research assistant. Your task is to retrieve the candidates that are highly likely to be cited by the query paper below. Query: <pre>{query_paper}</pre>
Candidate	<i>(no instruction prefix)</i>

Table 7: QWEN3-EMBEDDING-8B instruction prefix applied to the query side only.

F Qualitative Examples of Low-Scoring Human References

Table 9 lists eighteen representative human-curated references that receive a low semantic relevance score (0 or 20 on the 0–100 rubric). Examples are sampled to cover the recurring categories of non-

topical citation we observe across the 250-paper set: foundational architectures (Inception, ResNet (He et al., 2016), Llama-3, DDPM), classical algorithms (Adam (Kingma and Ba, 2015), CMA-ES, GELU, the Concrete distribution), software libraries (PyTorch, scikit-learn, CatBoost), benchmarks (ImageNet, DQN/Atari), and cross-domain references (autonomous-vehicle localization, asset pricing, OpenAlex). These citations are common and typically well-motivated in the body, yet their abstracts contain no topical signal that an abstract-conditioned judge can use.

G Artifact Licenses and Use

We use the following pre-existing artifacts in their published configurations: GPT-OSS-120B (Apache 2.0), QWEN3-EMBEDDING-8B (Zhang et al., 2025a) (Apache 2.0), vLLM (Kwon et al.,

Role	Content
System	You are an expert academic research assistant. You will be shown a query paper and a candidate paper and your task is to analyze the semantic relevance of the query paper and the candidate paper. Assess how relevant the candidate paper is to the subject matter, research scope, and focus of the query paper. Consider topical overlap, methodological similarity, shared objectives, and whether the candidate contributes meaningfully to the themes of the query paper.
User	You are an expert academic research assistant. Input. Query Paper Details: Title: {query_title} Abstract: {query_abstract} Full Paper: {query_full} Candidate Paper: Title: {candidate_title} Abstract: {candidate_abstract} Instructions. Analyze the semantic relevance of the query paper and the candidate paper. Assess how relevant the candidate paper is to the subject matter, research scope, and focus of the query paper. Task rubric. • 5 (Direct Correspondence): Candidate directly addresses the same research problem as the query paper. • 4 (Primary Topical Focus): Candidate’s central theme is closely related to the query paper. • 3 (Substantial Topical Coverage): Candidate covers significant aspects of the query paper’s domain. • 2 (Peripheral Topical Treatment): Candidate addresses the query paper’s subject as a secondary element. • 1 (Tangential Relevance): Minimal substantive overlap. • 0 (No Substantive Relevance): Candidate is from a different domain or research area. Output format. <pre>{ "paper_to_paper_relevance": { "relevanceScore": 0, "confidenceLevel": 0, "summaryStatement": "... " } }</pre>

Table 8: Semantic relevance judge prompt. Returns an integer 0–5 per (query, candidate) pair; we multiply by 20 to obtain the 0–100 scale used throughout the paper.

2023) (Apache 2.0), and HDBSCAN (McInnes et al., 2017) (BSD-3). Reference data comes from OpenAlex (Priem et al., 2022) (CC0), Semantic Scholar (Kinney et al., 2023) (standard API terms of service), and arXiv preprints (per-author licenses selected at submission). All uses align with the research purposes the source artifacts permit. The artifacts we will release (the Deep Research pipeline code, the LLM-judge prompts, the OpenAlex distance-analysis scripts, and ROLLINGEVAL-JUN25) will be made available under Apache 2.0 for code and CC-BY-4.0 for the benchmark.

Cited work	Cited topic	Citing paper’s focus	Judge rationale (abridged)	Score
CMA-ES tutorial (1604.00772)	Continuous black-box optimization	Unsupervised environment design for RL	“Continuous black-box optimization; does not address unsupervised environment design or curriculum learning for RL.”	0
Inception (1512.00567)	Convolutional vision architecture	Multi-perspective disagreement-aware NLP modeling	“Computer vision architecture improvements; no overlap with multi-perspective, disagreement-aware NLP modeling.”	0
Concrete distribution (1611.00712)	Continuous relaxation of discrete random variables	Heterogeneous graph contrastive learning	“Continuous relaxations of discrete random variables; unrelated to heterogeneous graph contrastive learning.”	0
DDPM (2006.11239)	Diffusion probabilistic image synthesis	Volumetric neural body modeling and collision handling	“Diffusion probabilistic image synthesis; unrelated to volumetric neural body modeling, contacts, and collision handling.”	0
No-norm Transformers (2503.10622)	Eliminating normalization in Transformers	Test-time adaptation for time-series forecasting	“Eliminating normalization in Transformers; unrelated to test-time adaptation or time-series forecasting.”	0
OpenAlex (2301.10140)	Open scholarly data platform and knowledge graph	Multi-agent LLMs for language model architecture discovery	“Open scholarly data platform and knowledge graph; unrelated to multi-agent LLM architecture discovery.”	0
Asset-pricing ML (2403.06779)	Machine learning in asset pricing (finance)	Multi-agent LLMs for language model architecture discovery	“Machine learning applications in asset pricing (finance); unrelated to autonomous LM architecture discovery.”	0
AV localization (1906.01061)	Autonomous-vehicle localization requirements	LLM-powered multimodal accessibility agents	“Autonomous vehicle localization requirements; unrelated to LLM-powered multimodal accessibility agents.”	0
Llama-3 (2407.21783)	Foundation language model and capabilities	Attention-logit analysis and contextual compression	“Describes a new LLM and its capabilities; lacks focus on attention-logit analysis or compression/attribution methods.”	20
CatBoost (1810.11363)	Gradient-boosting library for categorical features	Multi-agent multimodal accessibility	“Gradient-boosting library for categorical features; only tangential overlap in generic ML methods.”	20
Adam (1412.6980)	Stochastic optimization for deep learning	Orthogonal finetuning with matrix-free computation	“Generic stochastic optimization method; does not address orthogonal finetuning or matrix-free computation.”	20
ResNet (1512.03385)	Deep residual networks for image classification	Diffusion inference-time alignment via tree search	“ResNet for image classification; does not address diffusion inference-time alignment or tree-search sampling.”	20
GELU (1606.08415)	Activation function for neural networks	Diffusion inference-time alignment via tree search	“Introduces an activation function; only a peripheral connection through the broader deep-learning context.”	20
Sentence-BERT (1908.10084)	Sentence embedding methods	Evaluation of LLM-generated research ideas	“Sentence embedding methods; only a tangential connection to LLM ideation-execution evaluation.”	20
PyTorch (1912.01703)	Deep-learning library implementation	LLM safety and long-horizon risk simulation	“Deep-learning library implementation; does not address LLM safety, alignment, or long-horizon risk simulation.”	20
scikit-learn (1201.0490)	General-purpose machine-learning library	Hierarchical concept-bottleneck reasoning	“General-purpose ML library; does not address hierarchical concept reasoning or graph-based CBMs.”	20
ImageNet ILSVRC (1409.0575)	/ Large-scale image classification benchmark	Surgical phase recognition with state-space models	“Generic image classification benchmark; does not address surgical phase recognition or video modeling.”	20
DQN (Atari) (1312.5602)	Value-based deep reinforcement learning	Hierarchical planning for long-horizon goal-conditioned tasks	“Value-based deep RL on Atari; does not address hierarchical planning or long-horizon goal-conditioned tasks.”	20

Table 9: Example low-scoring human references with the judge’s abridged rationale. Score is on the 0–100 rubric. All examples drawn from the 250-paper human ground-truth set used in Section 5.